

AI-DRIVEN PROGRESS EVALUATION AND RISK DETECTION

Project ID: 25-26J-374

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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Declaration

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I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

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The supervisors should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

S. S. I. R. K.

Signature of the supervisor

(Prof. Samantha Rajapaksha)

29/8/25

Date

Abstract

The success of academic projects depends heavily on effective supervision, yet many institutions still use labor-intensive, arbitrary, and mismatch-prone manual allocation techniques. This project suggests creating and implementing a supervisor-student matching and NLP-based mentoring communication system in order to overcome this difficulty. By analyzing project subjects and contrasting them with supervisor availability and experience, the main goal is to automate the process of matching students with supervisors. This method guarantees a more equitable and precise distribution of supervisory resources while lowering administrative burden. To extract keywords, preprocess given subjects, and provide semantic representations that accurately convey their intended meaning, the system uses Natural Language Processing algorithms. A ranked list of the best supervisors for each project is produced by comparing these representations with supervisor profiles to determine similarity scores. To maintain productive contact, the system offers a dedicated communication channel in addition to allocation. To help students with common project-related questions and clarifications, a chatbot built on neural networks is incorporated, cutting down on information retrieval delays. Through the platform, supervisors continue to have direct interactions with students for academic advice and mentoring; the chatbot supports this process by answering general or commonly asked questions. In order to handle requests, matching logic, and the chat interface, the platform will be created as a web application. To improve transparency and guarantee more seamless supervisory interactions, descriptive metrics like message frequency and response times will be monitored. By combining intelligent allocation with structured communication support, the system is expected to improve supervisor–student alignment, enhance efficiency, and contribute to more effective project management in academic environments.

Keywords: Academic progress monitoring, predictive analytics, LSTM forecasting, sentiment analysis, risk evaluation, intelligent dashboards, real-time supervision, machine learning in education, project performance tracking.

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1. Introduction

Ensuring continuous project progress tracking and early risk detection is one of the most important difficulties encountered by supervisors and students in higher education, especially when it comes to final-year academic projects. Although supervisors are essential in helping students, manual reporting and subjective evaluation have been the mainstays of project milestone monitoring in the past. This method frequently leads to delays, insufficient feedback loops, and a lack of insight into students' participation in real time, which hinders timely interventions and has a detrimental effect on project results. [1, 2]

Research has highlighted the value of proactive risk identification and structured assessment in academic projects because they allow supervisors to not only evaluate performance but also foresee and address any problems before they become more serious.[3]. However, the intricate interdependencies of contemporary academic projects make traditional approaches like handwritten progress reports, ad hoc feedback sessions, and periodic evaluations inadequate. These manual procedures lead to missing hazards, inconsistent evaluations, and less accountability as project groups get bigger and more complicated.[4].

The combination of machine learning (ML) and artificial intelligence (AI) approaches has become a viable approach to filling these gaps. Long Short-Term Memory (LSTM) networks are particularly useful for project risk and performance deviation predictions because they can be used to represent sequential data, including team activity logs, communication histories, and submission patterns [5]. Simultaneously, the system can identify hidden patterns of stress or disengagement that are otherwise hard to measure thanks to sentiment analysis of supervisor comments and student contacts[6] [7]. Supervisors can get timely notifications for remedial action by combining these insights with anomaly detection techniques to indicate abrupt departures in productivity or cooperation patterns [8].

The combination of machine learning (ML) and artificial intelligence (AI) approaches has become a viable approach to filling these gaps. Long Short-Term Memory (LSTM) networks are particularly useful for project risk and performance deviation predictions [9] [10].

By facilitating evidence-based assessment, proactive intervention, and predictive monitoring, this component ultimately seeks to improve the project supervision process as a whole. It is in line with current research directions in educational data mining and learning analytics, which support the use of AI-based risk detection, multimodal learning signals, and temporal data to improve student outcomes [11] [12] [13]. With this strategy, the system hopes to help create an academic project ecosystem that is more open, accountable, and effective, where students are assisted in more successfully accomplishing their goals and supervisors are equipped with useful insights.

2. Background & Literature survey

2.1 Background

Academic projects, especially final-year group projects, are essential for evaluating students' learning, teamwork, and problem-solving skills in higher education. Nonetheless, a recurring issue in project-based learning settings is the absence of organized systems for tracking advancement and identifying hazards early. Manual supervision, milestone reviews, and ad hoc reporting are the mainstays of traditional approaches, which are frequently laborious, subjective, and have a limited capacity to record changes in team performance in real time [1] [2]. Since project groups usually entail a number of interdependent tasks, missed deadlines, inadequate communication, or unequal participation frequently go unnoticed until later on. This leads to a lower-quality project, unhappy students, and needless academic stress.

Supervisors are finding it more and more difficult to effectively monitor diverse teams as a result of the growth of higher education and the increase in project-based assessments [2]. Inefficiencies arise when traditional supervision techniques, like planned progress meetings and static reporting, are the only ones used. Instead of offering a continuous assessment, these techniques only offer a moment in time of performance. As a result, minor hazards like waning motivation, inadequate teamwork, or communication breakdowns are usually disregarded until they become serious project failures.

This problem highlights the pressing need for automated and data-driven systems that can track student activities, offer unbiased assessments, and identify hazards in real time [3] [5]. Learning analytics (LA) and educational data mining (EDM) developments have brought methodical approaches to gathering, evaluating, and interpreting student data to aid in higher education decision-making [1] [2]. These disciplines place a strong emphasis on modeling student behavior and performance trends by using temporal patterns in data, such as activity logs, assignment submissions, and communication

traces. Time-series data analysis in particular has become crucial for identifying behavioral changes that could indicate project risks or productivity gaps [1].

Long Short-Term Memory (LSTM) networks have become increasingly effective tools for modeling sequential data and forecasting performance outcomes in recent years. Because LSTMs can capture long-range dependencies and spot patterns that might not be apparent through manual observation, they are a good fit for handling time-dependent project data, like weekly task updates or communication logs, in contrast to traditional machine learning models [3] [4]. LSTM-based models have been successfully used in academic risk identification and student performance prediction in the past, showing promise in predicting possible dropouts, grade declines, or participation problems [5]. Simultaneously, sentiment analysis and opinion mining have grown in significance in educational settings. Sentiment analysis methods can reveal team morale, engagement levels, and possible conflicts by examining communication logs (emails, chat messages, and status updates) [8]. In project-based learning, where collaboration quality is just as crucial as technical output, these methods are especially beneficial. Research indicates that unfavorable sentiment patterns or sudden changes in sentiment frequently precede problems with productivity, enabling managers to take early action.

Furthermore, proactive interventions can be greatly improved by incorporating risk detection models into academic project environments. Predictive models can detect risk factors like delays, workload imbalances, and unmet requirements, as demonstrated by earlier works in software project management and agile environments [12] [13]. By applying these strategies in academic settings, supervisors can minimize failures and guarantee equitable evaluation procedures by receiving real-time alerts about team anomalies, such as persistently missed deadlines or low participation.

Lastly, dashboards have shown success in both the professional and educational spheres for visualizing risks and progress. Transparency, accountability, and prompt remedial action are made possible by dashboards, which give supervisors and students clear, real-time feedback [14]. This is in line with more general learning analytics trends, which

prioritize ongoing observation, insights derived from data, and interventions supported by evidence to enhance learning outcomes [15] [16].

In conclusion, this study's background is situated at the nexus of learning analytics, predictive modeling, and educational data mining. The size and complexity of contemporary academic projects are too great for traditional supervision techniques, but AI-driven strategies that integrate sentiment analysis, anomaly detection, and LSTM-based forecasting offer a complete answer. With the help of these technologies, project supervision could become a more proactive, open, and equitable process, which would improve student outcomes and lower the likelihood of subpar projects.

2.2 Literature review

2.2.1 Domain Analysis

Artificial intelligence (AI) and machine learning (ML) techniques have become more prevalent in academic project management in recent years. These techniques are used to track student progress, identify potential hazards, and assist supervisors in making sure that projects are completed on time. Conventional methods of tracking progress depended on supervisor feedback and manual evaluations, which were frequently arbitrary, unreliable, and unable to provide real-time insights into student performance. Automated approaches have become a key focus in educational data mining and learning analytics due to the increasing complexity of higher education projects and the growing need for timely interventions [1] [2].

The use of Long Short-Term Memory (LSTM) models, which have demonstrated efficacy in predicting time-series data, especially trends in student performance, is one of the major developments in this area [3] [4]. Sequential dependencies in student activity logs, submission histories, and communication patterns can be captured by LSTM networks, allowing predictive analytics to identify possible hazards before they become apparent. According to studies, supervisors can take prompt corrective action when LSTM is integrated with academic performance data, providing early warnings about underperformance [5].

Sentiment analysis has become a useful technique for assessing group communication quality and student engagement in addition to time-series modeling. Supervisors can gain insights into student motivation, collaboration dynamics, and possible conflicts by using sentiment analysis techniques to examine text-based communication, such as chat logs, discussion threads, and progress reports. [8] According to recent surveys, sentiment analysis is especially helpful in educational settings for identifying subtle behavioral cues that manual reviews would otherwise overlook. [17].

Additionally, studies on risk detection frameworks in the software project and educational domains have demonstrated the accuracy of deep learning and anomaly

detection methods in detecting behavioral abnormalities, irregular participation, and productivity gaps [12] [13]. Supervisors can monitor project health in real time with these systems, which combine predictive models with dashboards and early-warning alerts. In addition to increasing transparency, these frameworks enable supervisors to take a proactive approach to intervention rather than depending solely on retrospective assessments.

Lastly, learning analytics platforms and real-time dashboards have emerged as crucial resources for giving managers useful information. Research indicates that dashboards improve decision-making by combining various data streams, including communication logs, deadlines, and performance metrics, into visually represented trends.[14] [18] Dashboards can help supervisors identify at-risk teams, provide timely feedback, and promote better project outcomes when paired with predictive models and risk detection systems.

In general, the field demonstrates a distinct shift from manual and retroactive assessment techniques to AI-powered real-time intervention and predictive monitoring systems. This change is in line with higher education institutions' overarching objectives to improve project supervision's efficiency, accountability, and transparency [15].

2.2.2 Technical Review

The AI-Driven Progress Evaluation and Risk Detection component is based on sophisticated deep learning architectures that can efficiently handle temporal and sequential data. Because LSTM networks can capture long-range dependencies in sequential datasets, they have become one of the most popular methods for time-series forecasting in education and learning analytics [3] [4]. In contrast to conventional recurrent neural networks, LSTMs overcome the issue of vanishing gradients, which allows them to accurately model lengthy project timelines and activity sequences. Because of this, they are especially well-suited for assessing the progress of academic projects, which frequently involve interdependent milestones and changing student behaviors.

In educational settings, recent uses of LSTM-based models have demonstrated encouraging outcomes in terms of risk identification, student performance prediction, and early intervention support [5] [9]. Empirical research, for instance, shows that LSTMs can predict possible delays or poor performance by processing continuous streams of student activity data, including submissions, attendance at meetings, and communication patterns. Academic project monitoring systems can offer predictive insights by implementing such models, guaranteeing administrators and supervisors take proactive rather than reactive action.

Sentiment analysis has been extensively incorporated into educational analytics to capture the behavioral and emotional aspects of student engagement, in addition to LSTM forecasting. Based on communication logs like chats, reports, or feedback exchanges, surveys of sentiment analysis algorithms show how well they can identify productivity gaps, disengagement, and possible hazards [8] [17]. By providing contextual understanding of textual interactions, neural models enhanced with attention mechanisms—like transformers—further enhance this capability and enable a more nuanced interpretation of team morale and collaborative dynamics [6] [7].

This component's technical focus also includes anomaly detection, which is crucial for spotting departures from typical activity patterns. Unusual behaviors like missing deadlines, abrupt drops in activity, or unusual communication patterns can be effectively identified using deep learning techniques like autoencoders and hybrid architectures [12] [13]. These techniques offer a safeguard that guarantees problems are found early and dealt with right away, lowering project risks. Deep learning-based anomaly detection has been shown to reduce failure rates by continuously adapting to new patterns in dynamic environments, according to studies on risk prediction for software and academic projects [13] [18].

Furthermore, real-time visualization of performance metrics and risk alerts is made easier by the integration of AI-driven dashboards. To give supervisors useful information, these dashboards combine sentiment analysis outputs, anomaly detection

results, and predictions from LSTM models [14]. Technically speaking, this strategy is in line with contemporary real-time learning analytics systems, which prioritize openness, interpretability, and proactive monitoring for academic achievement.

All things considered, this component's technical underpinnings are a combination of strong anomaly detection techniques, sentiment analysis for semantic understanding, and sequential modeling using LSTMs. When combined, these technologies allow for evidence-based decision-making, proactive intervention, and predictive monitoring in academic project management. The system reduces the possibility of project risks and facilitates sustainable progress tracking by combining these techniques into a single framework, guaranteeing a trustworthy academic supervision environment.

2.2.3 Similar Systems Study

1. Brightspace by Desire2Learn (D2L)

A popular learning management system (LMS) in higher education, Brightspace gives teachers and students access to tools for managing assignments, tracking engagement, and providing feedback. It uses data-driven dashboards to monitor course progress, assignment submissions, and student involvement [14]. However, rather than offering predictive insights into possible risks or anomalies, its primary focus is on descriptive analytics, which highlight what has already occurred. Although engagement analytics are supported by the platform, it is devoid of tools for early risk identification in project groups or real-time performance trend forecasting. This reduces its usefulness for tracking academic projects, where early intervention is essential.

2. Microsoft Teams (Education Version)

Microsoft Teams, which provides real-time communication via chat, video conferencing, and file sharing, has become popular as a collaboration center in academic settings. It makes group-based project work convenient by facilitating smooth communication between supervisors and students. Despite these advantages, Teams lacks built-in tools for monitoring student progress or anticipating potential problems

with workload imbalances, delays, or productivity gaps at the team level [14]. Its general functionality emphasizes teamwork over domain-specific predictive analytics. As a result, although Teams improves communication effectiveness, it is unable to predict academic risks or identify underperforming teams on its own.

3. Moodle + Analytics Plugins

Plugins that create performance dashboards and give teachers information about student activity patterns have been added to Moodle, another popular LMS [14]. For descriptive monitoring, such as identifying students who have not accessed resources or turned in assignments on time, these dashboards work well. However, Moodle's analytics are still primarily reactive and retrospective, lacking any predictive tools to identify possible hazards before they materialize. Additionally, rather than offering group-level predictive evaluation that is necessary for final-year project management scenarios, its scope is primarily individualized—focused on student–instructor interactions.

4. LSTM-Based Student Performance Prediction Systems

The application of Long Short-Term Memory (LSTM) models to forecast student academic performance has been investigated recently [5]. These systems show increased accuracy in capturing temporal dependencies when time-series forecasting is applied to academic data, such as grades, attendance, or test scores. Even though these models are good at forecasting future academic results, they are frequently only used for individual-level forecasting rather than cooperative group projects. Furthermore, the majority of implementations are less appropriate for adaptive academic project management, where progress dynamics change frequently, because they do not integrate with real-time project monitoring environments.

5. Sentiment Analysis and Risk Dashboards

Sentiment analysis has been used to assess user interactions, communication logs, and engagement patterns in social and educational contexts [17]. NLP-powered risk dashboards can identify possible dropout risks, flag disengagement, and highlight

irregularities in student communication. Despite showing promise, these methods are typically developed for large-scale social media analysis or as standalone research prototypes rather than academic project settings. Furthermore, the majority of sentiment-driven systems are not integrated with predictive forecasting models such as LSTM, which prevents them from offering a comprehensive perspective of academic teams' performance trends and behavioral signals. [18]

2.2.4 Comparative Analysis

All of the systems under review provide some partial answers to academic monitoring and supervision. Although platforms like Moodle and Brightspace offer descriptive dashboards, their inability to make predictions limits proactive intervention. Although Microsoft Teams is excellent at communicating, it is unable to provide organized project risk monitoring or assessment. Although LSTM-based models [5] improve performance prediction, they ignore group-level interdependencies that are crucial for final-year projects in favor of individual outcomes. In the meantime, sentiment analysis dashboards [17] [18] identify patterns of engagement but are incomplete and disjointed if they are not integrated into predictive frameworks.

The suggested AI-Driven Progress Evaluation and Risk Detection component, on the other hand, integrates sentiment analysis, anomaly detection, and LSTM forecasting into a single system designed specifically for academic projects. In addition to assessing present progress, this integration makes it possible to forecast possible hazards like missed deadlines, unequal team contributions, and decreases in productivity. Additionally, the system will show performance dashboards and evaluation scores, providing supervisors and students with transparent accountability as well as predictive insights. By offering a thorough, proactive, and academically specific monitoring framework, this all-encompassing design directly addresses the shortcomings in the current systems.

2.3 Research Gap

Table 1: Research Gap

Existing system	Shortcomings	Proposed Solution
SpringerOpen	No predictive or proactive monitoring	Integrates predictive LSTM forecasting to provide early warnings rather than just past data summaries
SpringerLink	Analytical methods used separately	Combines sentiment analysis with forecasting and anomaly detection for comprehensive, integrated risk evaluation.
MDPI	Absence of real-time dashboards and early intervention tools	Provides real-time dashboards with live risk alerts and progress insights for proactive interventions.
Ojed	Reliance on manual reporting and lack of structured supervision tools	Automates monitoring through LSTM forecasts & risk detection, reducing manual workload and increasing consistency.

2.4 Research Problem

In higher education, monitoring and evaluating student project groups effectively is still a major challenge, particularly in large-scale settings where multiple teams are working on interdependent tasks. Institutions frequently use static progress reports, manual grading, and periodic reviews, but these methods are reactive and ineffective because they don't account for the dynamic nature of team activities, risks, and productivity in real time. Lower performance, missed deadlines, and compromised results result from supervisors' inability to recognize troubled teams early, address hidden risks, and ensure prompt interventions.

The majority of current systems, like learning management platforms, only offer engagement tracking or descriptive analytics; they don't have any specialized features for anomaly detection, predictive monitoring, or real-time evaluation dashboards. Supervisors are consequently left with disjointed data and delayed insights, as well as no tools for predicting failures or identifying problems like low participation or unequal contributions. In final-year projects, where prompt risk identification is crucial, the gap is most noticeable. Although some studies have used sentiment analysis to identify engagement and machine learning models like LSTM to predict performance, these initiatives are still separate and not incorporated into a comprehensive monitoring framework.

The lack of a comprehensive system that integrates sentiment analysis of team communications, LSTM-based progress forecasting, and real-time dashboards for proactive risk detection is therefore the main research issue. In the absence of it, organizations continue to rely on retrospective assessments, which restricts early interventions and lowers project success rates overall.

2.5 Objectives

2.5.1 Main Objectives

To design and develop an AI-Driven Progress Evaluation and Risk Detection System that leverages LSTM-based predictive modelling, sentiment analysis, and anomaly detection to monitor academic project groups, forecast risks, and provide actionable insights for timely interventions and improved project outcomes

2.5.2 Specific Objectives

1. Progress Forecasting Using LSTM

To implement Long Short-Term Memory (LSTM) models for time-series forecasting of project performance trends.

Specific: Focused on forecasting progress and detecting delays in project timelines.

Measurable: Progress evaluated through forecast accuracy (e.g., RMSE, MAPE).

Achievable: Implementable using available project log and evaluation data.

Realistic: Supports monitoring within the scope of academic projects.

Time-bound: To be developed, trained, and tested within the project cycle duration.

2. Sentiment Analysis of Team Communication

To apply sentiment analysis on project communication logs (chat, reports, feedback) to measure engagement, collaboration, and motivation levels.

Specific: Targets real-time detection of student engagement trends.

Measurable: Engagement measured through sentiment polarity scores and classification accuracy.

Achievable: NLP libraries and pre-trained models can be adapted for project data.

Realistic: Feasible within academic project constraints.

3. Anomaly Detection for Risk Identification

To integrate anomaly detection methods that automatically flag unusual activity patterns, productivity drops, or irregular performance.

Specific: Identifies risks by monitoring abnormal behavior.

Measurable: Risks measured by anomaly detection precision and recall scores.

Achievable: Achieved through statistical and ML-based anomaly detection techniques.

Realistic: Applicable with structured student activity and progress data.

Time-bound: Integrated and tested within the system prototype timeline.

4. Real-Time Performance Dashboards

To design dashboards that visualize team progress, risk alerts, evaluation marks, and predictive insights in real time.

Specific: Provides supervisors and students with real-time visual insights.

Measurable: Dashboards evaluated based on usability testing and feedback scores.

Achievable: Supported through existing visualization frameworks

Realistic: Scalable for academic use without high resource demands.

Time-bound: Deployed in the final implementation stage of the project.

5. System Effectiveness Evaluation:

To evaluate the impact of the system in enhancing project supervision, reducing risks, and improving academic outcomes.

Specific: Assesses improvements in risk detection and project performance.

Measurable: Compared against baseline metrics (e.g., project delays, risk incidents).

Achievable: Based on experimental results with supervised project groups.

Realistic: Addresses actual challenges in academic project monitoring.

Time-bound: Final evaluation conducted at the end of the development cycle.

3. Methodology

3.1 System Overview

Predictive analytics and anomaly detection can give supervisors ongoing insight into student performance, and it is suggested that an AI-driven system for assessing project progress and identifying risks be put into place. The system combines two main components: an NLP-based sentiment analysis engine with anomaly detection to track communication and spot early warning indicators of dangers, and a Long Short-Term Memory (LSTM) network for time-series forecasting of project progress.

The LSTM forecasting model analyzes past trends to forecast future performance trajectories after receiving project activity logs, evaluation results, and communication data. Supervisors and students are able to foresee possible delays or gaps in productivity well in advance thanks to this. The sentiment analysis module assesses team communications concurrently, looking for indications of disengagement, disagreements, or low motivation. To further strengthen risk identification, an anomaly detection layer is then used to highlight odd performance trends or departures from expected norms.

Interactive performance dashboards are used to display the findings of these analyses. These dashboards show real-time sentiment trends, risk alerts, evaluation marks, and anticipated progress. While students get a better understanding of their own development and areas for growth, supervisors are able to monitor several projects at once. The system lessens reliance on manual oversight by automating the evaluation and risk detection process, freeing supervisors to concentrate on strategic mentoring and remedial actions rather than routine monitoring [15].

Projects are not only efficiently tracked but also protected against possible failures thanks to this dual structure, which combines proactive risk detection with predictive progress evaluation. By combining sentiment analysis, anomaly detection, and LSTM forecasting, a data-driven approach to project supervision is promoted, improving productivity, accountability, and general academic results.

3.1.1 System diagram

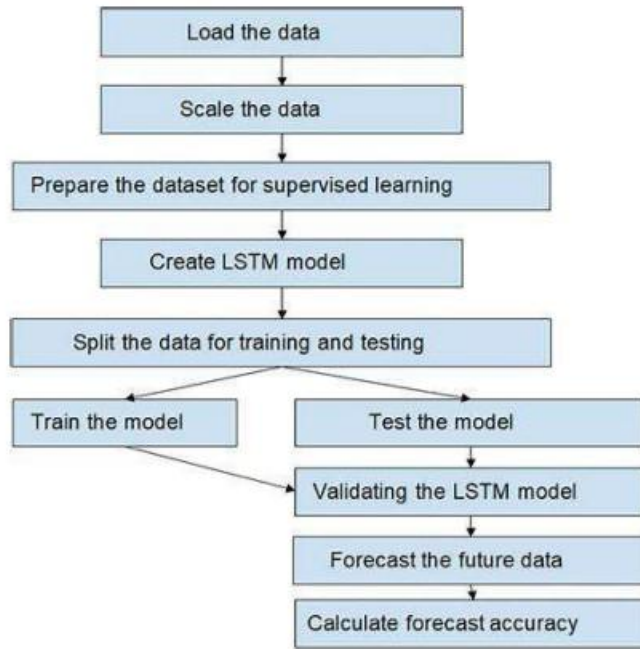


Figure 1: System diagram

The LSTM-based progress evaluation and risk detection model's sequential implementation is depicted in the system diagram. To guarantee uniformity and reduce bias among various variables, student project data is first loaded and scaled. The model is then able to efficiently map input features with corresponding outcomes by preparing the dataset in a supervised learning format. To capture temporal patterns and sequential dependencies in project activities, an LSTM model is developed. The model's predictive performance is then assessed by splitting the dataset into training and testing subsets. The model is used to predict future project outcomes after it has been trained and validated. In order to assess the system's dependability in tracking student development and identifying possible hazards, the forecasts' accuracy is finally computed.

3.1.2 System Architecture

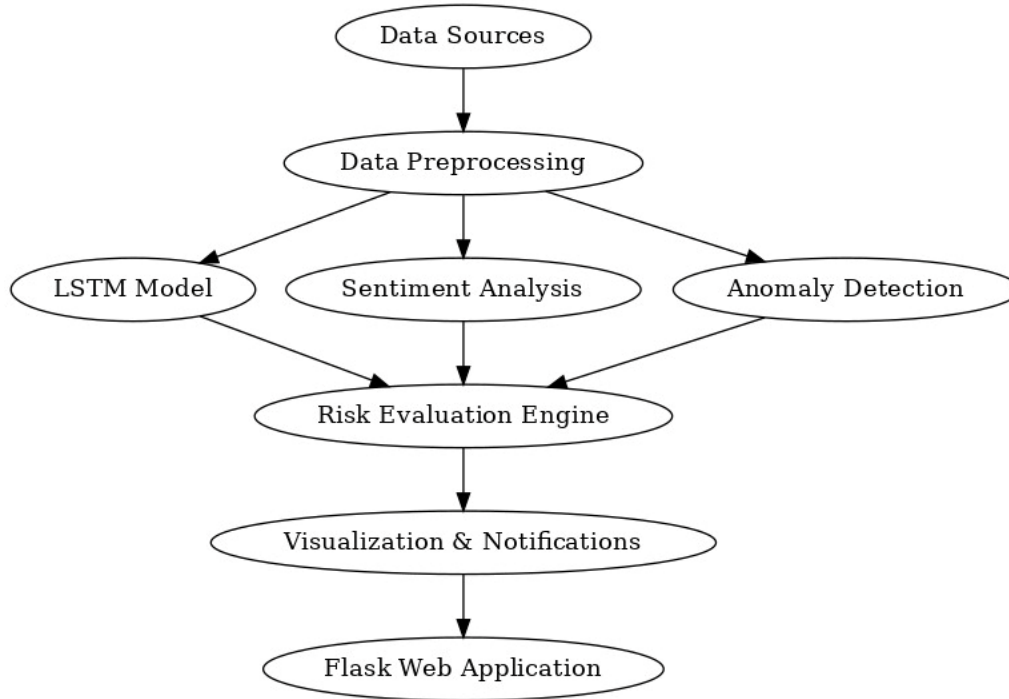


Figure 2: System Architecture

The system architecture describes how various elements intended for risk assessment and progress evaluation in academic projects are integrated. To guarantee quality and consistency, data from multiple sources is first gathered and then put through a preprocessing step. Three main modules are then used to analyze the processed data: anomaly detection to find odd behaviors or hazards, sentiment analysis to assess team communication, and an LSTM model to predict progress trends. A centralized Risk Evaluation Engine receives the outputs from these modules and compiles the information for use in making decisions. Actionable feedback is then provided by visualizing the assessed risks and progress indicators through interactive dashboards and notifications. The delivery platform is a Flask-based web application that allows students and supervisors to access the system instantly.

3.2 Technologies

3.2.1 Frontend Development

HTML5, CSS3, JavaScript, and React.js (for dynamic UI rendering) will be used in the development of the system frontend. Responsive design will be supported to guarantee accessibility on desktop, tablet, and mobile devices. This interface will be the main way that students and supervisors communicate. Predicted project progress, evaluation scores, anomaly alerts, and sentiment trends derived from the backend models will all be shown in interactive dashboards that are accessible through the frontend. Libraries like Chart.js and D3.js will be used to implement the visualizations, which will allow users to monitor performance over time and spot potential problems early. Supervisors will also be able to filter and compare team progress using a dedicated panel, and students will have access to their own unique performance insights.

The frontend will also include notification features to promote ongoing engagement, highlighting anticipated risks (like potential delays or dwindling engagement levels) and promptly alerting users for remedial action. By guaranteeing usability, interactivity, and real-time monitoring, this design will make the system useful and efficient in facilitating proactive intervention and bridging communication gaps.

3.2.2 Backend Development

Python Flask (or Django for scalability) will be used to implement the backend, which will offer a lightweight and modular REST-based architecture to manage communication between the analytical models and the frontend. In addition to managing data pipelines and coordinating machine learning tasks, this backend will serve as the main processing hub and deliver real-time responses.

Core backend functions will include:

- **LSTM Forecasting Module**
Processes historical project data and predicts future performance trends.
- **Sentiment Analysis Engine**
Applies NLP methods (e.g., transformers or RNN-based models) to evaluate student communications for motivation, stress, or disengagement signals.
- **Anomaly Detection Unit**
Flags irregularities in activity logs or evaluation scores
- **Data Integration Layer**
Handles secure storage, preprocessing, and retrieval of student activity logs, communication data, and evaluation marks.

To guarantee that updated forecasts and alerts are immediately displayed on the frontend, the backend will also manage real-time communication with the dashboards. The system will facilitate ongoing progress tracking, risk identification, and prompt decision-making for academic project supervision by offering a scalable and modular backend design.

3.2.3 Technical Pipeline for Progress Evaluation and Risk Detection

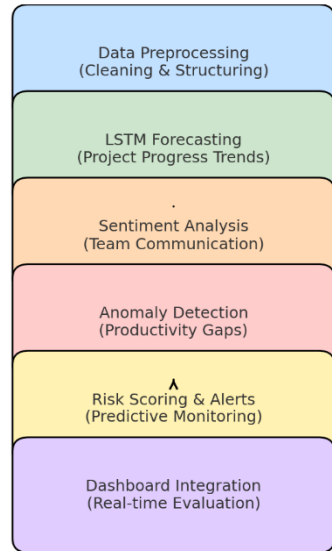


Figure 3:LSTM-Based Progress Evaluation & risk Detection Flow

The system will use a pipeline for predictive modeling and structured data processing to assess project team progress and identify risks. Data preprocessing is the first step in the workflow, where raw inputs like activity logs, reports that have been submitted, communication frequency, and evaluation scores are cleaned and normalized. To guarantee compatibility with deep learning models, this entails handling missing values, scaling numerical indicators such as participation frequency and submission timeliness and encoding categorical attributes.

Following preprocessing, the data will be converted into time-series sequences that show how team performance has changed over weeks or milestones. In order to capture temporal dependencies and long-term patterns in project activity, these sequences will be fed into Long Short-Term Memory (LSTM) networks [3], which are ideal for sequential data. By learning from past patterns, the LSTM will be trained to predict future performance trends, such as expected grades or the possibility of project delays.

Concurrently, a sentiment analysis module will use natural language processing (NLP) methods to process student communications, such as meeting notes, feedback logs, and discussion records [8] [17]. To provide an indirect indicator of team cohesion, motivation, and the quality of collaboration, this module will categorize text into three sentiment categories: positive, neutral, and negative. Sentiment signals, when paired with activity data, will enable the system to record qualitative engagement indicators in addition to quantitative performance metrics. After that, an anomaly detection engine will examine any departures from the norm, like abrupt declines in participation, erratic submission patterns, or spikes in negative sentiment. These irregularities will be noted on the supervisor dashboard and marked as possible hazards.

The pipeline's last step creates actionable insights by combining all of the results into a risk scoring framework. Early warning indicators (such as the likelihood of poor performance or disengagement) and visual projections of future results will be provided to supervisors and students. Proactive interventions to mitigate risks before they worsen are made possible by this predictive approach, which promotes prompt guidance and more dependable project outcomes.

3.2.4 LSTM-Based Progress Evaluation and Risk Detection

To provide real-time insights into project group performance and risks, the monitoring and evaluation module will use sentiment and anomaly detection techniques in conjunction with Long Short-Term Memory (LSTM) networks. This module will function as an intelligent monitoring layer and do the following:

Progress Forecasting: To produce forecasts of future performance trends, the LSTM model will process sequential project activity data, such as submissions, milestone completions, and group interactions. The system will be able to predict delays, spot performance lapses, and calculate the probability of meeting project milestones on schedule by utilizing historical data.

Sentiment and Engagement Analysis: The system will use sentiment analysis techniques to examine communication logs (such as meeting transcripts, discussions, or chat records) in order to capture the dynamics of group collaboration. This makes it possible to identify negative patterns that frequently precede productivity risks, like dissatisfaction, disengagement, or conflict.

Anomaly Detection: In order to identify anomalies in project activity, such as missed updates, unusual inactivity, or sudden declines in contribution levels, the module will incorporate anomaly detection algorithms. Supervisors are able to take proactive action because these anomalies serve as early warning indicators.

Integration with the Dashboard: The project monitoring dashboard will incorporate the findings from sentiment analysis, anomaly detection, and forecasting. In order to ensure prompt decision-making and remedial action, supervisors and students will have real-time access to risk scores, progress curves, and predictive alerts.

Fallback Mechanisms: The system will highlight the case for supervisors to manually review if the automated models produce predictions that are uncertain (such as low-confidence results). This guarantees that in academic project management, automated monitoring enhances rather than replaces expert judgment

3.3 Anticipated Conclusion

In order to improve proactive risk detection and progress evaluation in academic project groups, this recommendation suggests an AI-driven system. Supervisors and students can track performance trends, spot productivity gaps, and anticipate risks before they become serious with the help of the system, which uses LSTM-based time-series forecasting, sentiment analysis, and anomaly detection. By creating a predictive and data-driven monitoring channel, this strategy tackles contemporary issues like fragmented monitoring, delayed interventions, and subjective evaluations. A more organized, open, and proactive project management setting where teams are constantly led, risks are identified early, and managers can act more quickly and precisely is the anticipated result.

3.3.1 Results

It is anticipated that the provided system will increase the precision of risk assessments and performance evaluations in academic projects, in accordance with findings in the body of existing literature [5] [14] [18]. While anomaly detection techniques can draw attention to anomalies like missed milestones or abrupt declines in engagement, the integration of LSTM models is likely to yield accurate forecasting of progress trajectories. Furthermore, early identification of negative trends, stress, or student discontent will be made possible by sentiment analysis of communication data [17]. Instead of depending only on manual observation, supervisors will be able to respond quickly to data-driven insights by automating these processes. The suggested system will therefore encourage prompt interventions, lower the likelihood of project failure, and raise general student satisfaction and engagement. In the end, the system should result in improved project outcomes and increased academic project supervision efficiency.

3.3.2 Application

The system works best in academic settings where research projects and project-based learning call for ongoing oversight and direction. The system's predictive capabilities, which give supervisors early warnings of delays, risks, or decreased student engagement, can be advantageous to universities and other higher education institutions. While sentiment analysis uncovers hidden problems in team communication, LSTM-based forecasting and anomaly detection guarantee that managers can foresee possible project failures. This lessens the possibility of misunderstandings, enhances prompt interventions, and contributes to maintaining uniform academic standards among various project teams. Supervisors can focus on providing meaningful academic guidance and increase efficiency while improving learning outcomes by automating progress evaluation, which relieves them of the burden of excessive manual monitoring.

3.3.3 Real World Use

The suggested system is a useful tool for higher education institutions, such as colleges, universities, and training facilities, where supervisors oversee several students and projects at once. It is appropriate for large-scale academic settings due to its adaptable design, which enables it to be used across disciplines and academic levels, from undergraduate research to postgraduate studies. Beyond the classroom, professional mentoring and organizational training—where prompt performance insights are essential for success—can also benefit from the system's predictive risk detection and communication monitoring features. Through the integration of predictive analytics and intelligent communication tracking, the system helps managers, supervisors, and educators better match team or individual needs with guidance, making support proactive and contextually relevant.

3.4 Commercialization

Because the suggested system can be used for both professional project monitoring and academic supervision, it has a great deal of commercialization potential. The platform can be implemented on a subscription basis in universities and other higher education establishments to boost student outcomes, reduce project risks, and improve supervision effectiveness. Institutions lower administrative overhead while guaranteeing better supervisory resource allocation and prompt interventions by automating risk detection and progress evaluation.

The system can be commercialized for use in research organizations where project progress tracking is essential, corporate training programs, and internship management, in addition to academia. The risk-prediction and sentiment-analysis modules can be used by businesses, governmental organizations, and non-profits to identify early warning signs in project teams, ensuring proactive decision-making and lowering the chance of failures. This establishes the system as a flexible solution that can be applied to a range of work settings where risk assessment and ongoing performance evaluation are crucial.

Subscription-based pricing, institutional packages, and integration consulting services can all be used to generate revenue while maintaining sustainability and flexibility. The system can be customized for industry-specific uses like software engineering projects, healthcare training, or extensive research collaborations by presenting itself as a modular AI-driven platform. This commercial pathway offers scalable, intelligent risk detection and monitoring for academic and professional ecosystems, turning the initial academic research prototype into a long-term, marketable product.

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4.Gantt chart



Figure 4:Gantt chart

5. Work Breakdown System

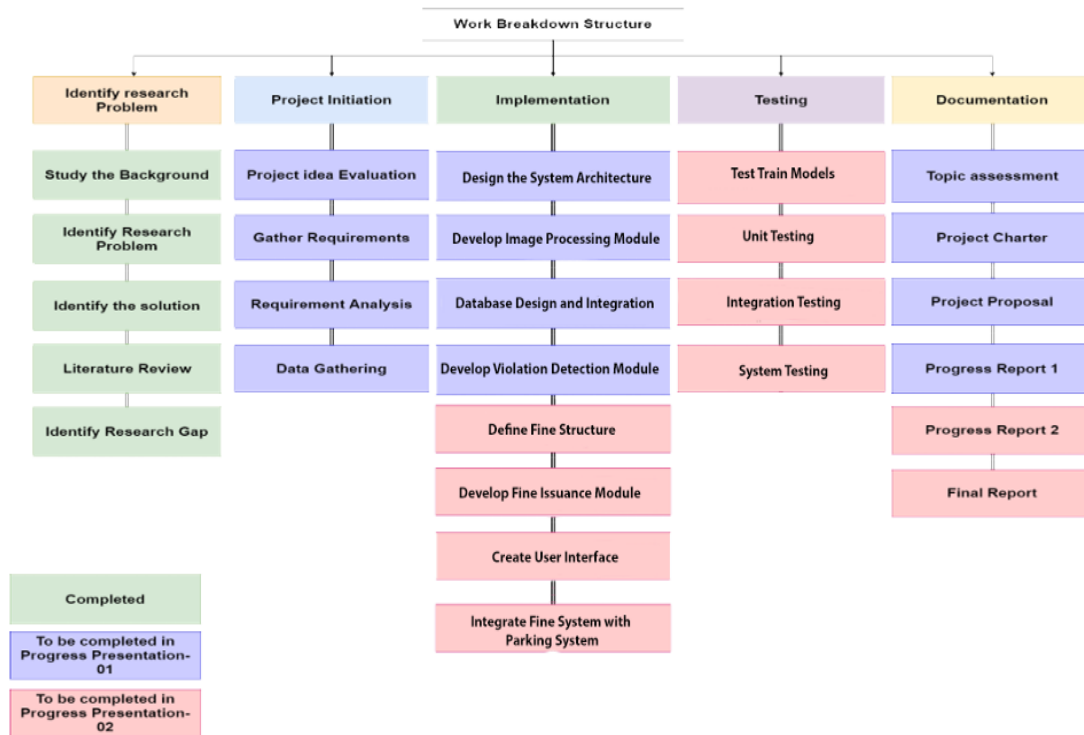


Figure 5: Work Breakdown System

6. Project Requirements

6.1 Functional Requirements

The system's necessary functions to accomplish the project's goals are specified by the functional requirements. The primary functions related to risk assessment, mentorship support, and progress evaluation are listed in the table below.

Table 2: Functional Requirements

No	Requirement Description	Priority
1	The system will allow students to submit periodic project progress updates (reports, milestones, deliverables).	High
2	The system will enable supervisors to register/update supervision capacity, workload, and specialization.	High
3	The system will process project updates and detect risks using NLP and machine learning models (e.g., sentiment analysis, anomaly detection).	High
4	A ranking and risk-prioritization mechanism will highlight projects requiring urgent attention.	High
5	The system will support real-time communication through an integrated chat interface.	High
6	The chatbot will handle project-related queries and provide automated risk alerts or guidance using neural-network-driven NLP.	Medium
7	The system will maintain detailed logs of communications, progress evaluations, and detected risks for future reference.	Medium

6.2 User Requirements

User requirements specify what various user types anticipate from the system. In terms of progress tracking, risk detection, access, interaction, and usability, these requirements take into account the needs of administrators, supervisors, and students. The system will guarantee a workable and efficient solution that complements the end users' workflow by concentrating on user expectations.

Table 3: User Requirements

NO	Requirement Description	User
1	Students will be able to log their project updates (tasks, milestones, deliverables) through a structured interface.	Student
2	Students will receive alerts and notifications when progress risks (delays, low performance indicators) are detected.	Student
3	Supervisors will be able to view real-time dashboards showing students' progress, milestones, and identified risks.	Supervisor
4	Supervisors will be notified automatically of high-risk cases (e.g., consistent delays, low engagement, missed deadlines).	Supervisor
5	Supervisors will be able to provide feedback and corrective actions through the integrated communication interface.	Supervisor
6	Administrators will monitor overall student performance analytics, system health, and supervisor load distribution.	Administrator
7	The system will include a secure login with role-based access (student, supervisor, administrator).	All Users
8	The system will automatically evaluate student activity data (submissions, logins, milestones) using AI models to flag at-risk students.	Student/ Supervisor

6.3 System Requirements

The operational and technical specifications necessary for the AI-driven progress evaluation and risk detection system to operate efficiently are outlined in the system requirements. These specifications guarantee that the platform can track student development, proactively detect hazards, and give managers useful information.

Table 4: System Requirements

NO	Requirement Description	System
1	The system shall provide a dynamic dashboard that visualizes project progress, milestones, and risk alerts in real time.	Software
2	Machine learning models (e.g., LSTM, Random Forest, or hybrid approaches) shall process historical and real-time student data to predict potential risks or delays.	Software
3	The backend will be developed in Python (Flask/Django) with integrated ML libraries.	Software
4	A rule-based decision layer shall complement the AI predictions to reduce false alarms and ensure explainability of detected risks.	Software
5	The system shall support role-based access (student, supervisor, admin) with personalized dashboards.	Security / Usability
6	The platform will log all progress evaluations, detected risks, and corrective actions for future auditing and research purposes.	Operational
7	The system shall support data import/export in CSV, Excel, and PDF formats for reporting and analysis.	Performance
8	The AI models will be regularly retrained with new project data to improve prediction accuracy and adapt to evolving academic patterns	Software
9	The system will provide automated early-warning notifications supervisors and students.	Communication
10	The platform shall be optimized for scalability to handle large datasets without performance degradation.	Performance

6.4 Budget and Budget Justification

Table 5 :Budget and Budget Justification

Item/Resource	Estimated Cost (LKR/USD)	Justification
Development Tools (IDE, ML Libraries, APIs)	30,000 LKR	Necessary for predictive model implementation, debugging, and coding. includes dashboard APIs, cloud-based model monitoring plugins, and licensed ML/DL libraries (TensorFlow, PyTorch, Scikit-learn).
Hosting & Deployment (Cloud/Server with GPU support)	75,000 LKR	Vital for implementing dashboards, risk assessment models, and backend services. For risk prediction tasks, GPU-enabled hosting guarantees effective training and inference in addition to round-the-clock system availability.
Hardware (Laptop/Workstation)	180,000 LKR	Processing progress data, training and testing machine learning, and rendering visualizations all require a powerful computer. necessary for keeping local backups and testing out various predictive models.
Data Storage & Processing	30,000 LKR	Training datasets, model checkpoints, and project progress logs are all kept in a separate location. supports retraining for increased accuracy and guarantees the safe handling of student and supervisor data.
Internet & Communication	15,000 LKR	Facilitates timely risk alert distribution via email and SMS notifications, remote supervisor dashboard access, and real-time collaboration.

Documentation & Reporting	12,000 LKR	Creating, printing, and binding dashboards, progress evaluation reports, and final project documentation.
Contingency (10%)	34,200 LKR	Set aside for unanticipated costs like extra cloud computing hours, storage expansion, unforeseen model retraining needs, or small hardware upgrades.

The AI-Driven Progress Evaluation and Risk Detection System's development and implementation are supported by the carefully crafted proposed budget. Since these are necessary for training and implementing machine learning models that will assess project progress and forecast risks in real time, a sizeable portion is devoted to hardware and cloud hosting with GPU support. Costs associated with data processing and storage guarantee the safe preservation of project updates, supervisor-student interaction logs, and historical datasets for ongoing model retraining and performance enhancement. Investing in ML libraries and development tools ensures access to the reliable frameworks needed to create interactive dashboards and predictive models.

Furthermore, communication costs will maintain supervisors' and students' real-time alerts and notifications, while documentation and reporting costs will pay for the creation of evaluation reports and the distribution of research findings. Lastly, a contingency reserve ensures system sustainability and dependability by offering flexibility to meet unforeseen needs like extra compute hours or scaling requirements.

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Appendix

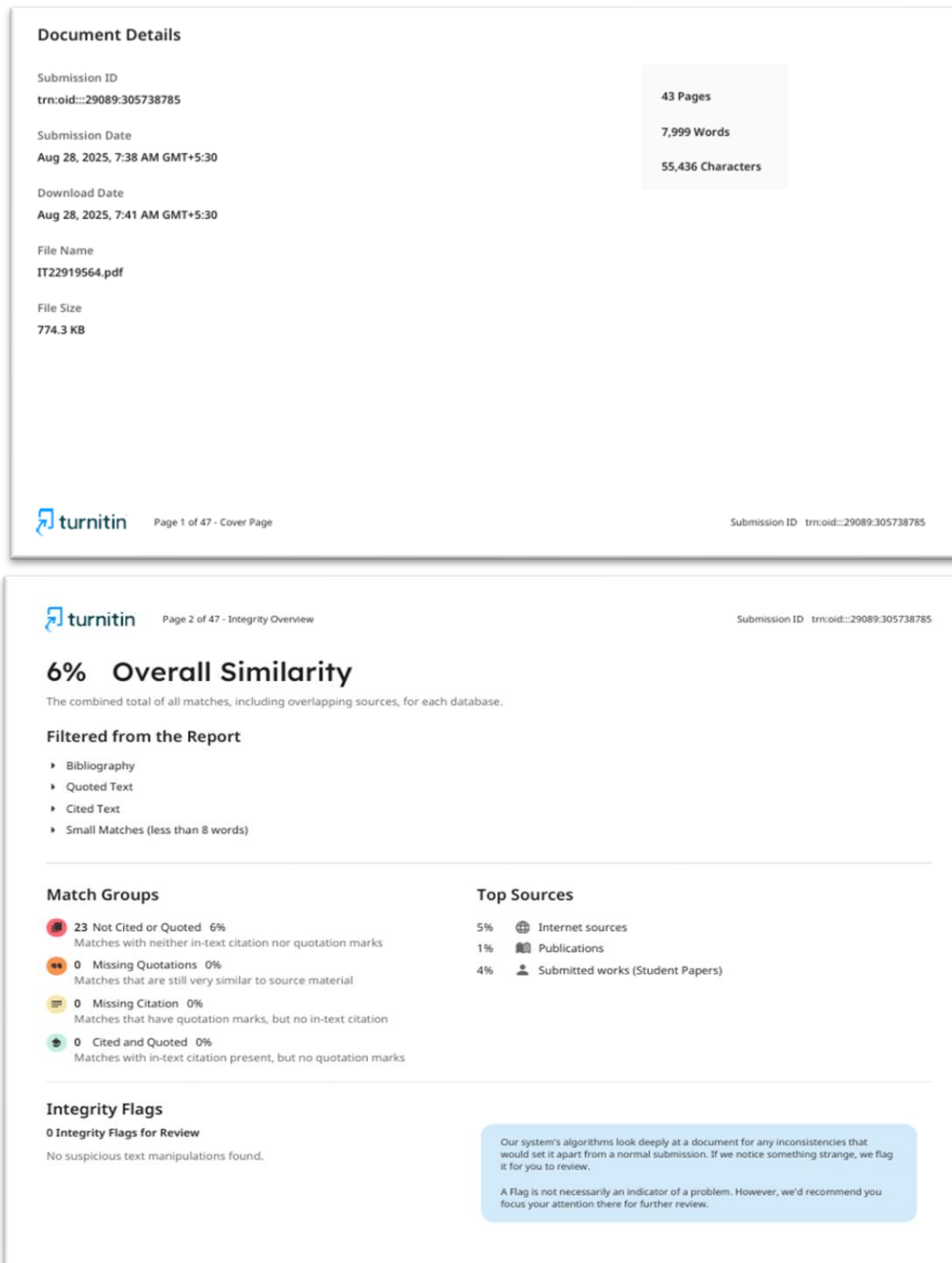


Figure 6 :Plagiarism Report